Recovered ECPP: independent runtime controls

2026-10-24 · Engineering observations · Named curve orders executed: **0**

The fixed ECPP V3.4.1 32-bit binaries now execute in fresh offline QEMU VMs. The report records bounded compatibility and failure controls. It identifies no common omitted-curve property or historical acceptance rule.

Recorded results

Check	Actual scope	Recorded result
Arithmetic	69 vectors, 1–521 bits; 5 exact comparisons each	345 pass
Prover	4 normal inputs; CM-only 101; empty/missing catalog	7 expected behaviors
Checker	5 positive and 4 negative fixtures	9 expected outcomes
Binding	9 saved checker cases + 7 injected failures	16 pass
Framing	17 saved serial transcripts + 9 corruptions	26 pass
Named cohort	11 omitted + 10 published exact UIDs/orders pinned	0 executed

Binding and framing reuse saved outputs. These counts are not independent curve samples. The arithmetic-only bring-up qemu-v2 and full controls qemu-v3 use the same 69 fixed vectors.

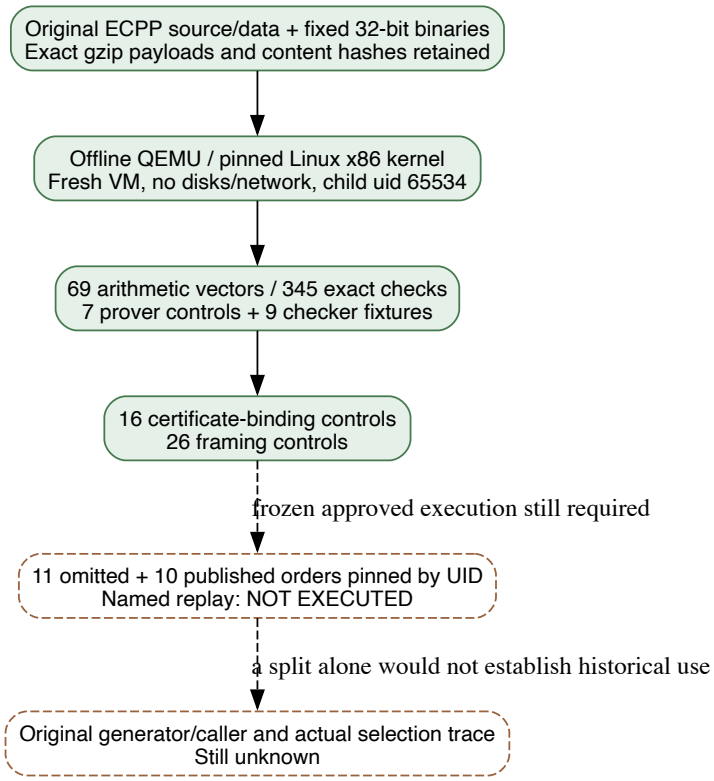


Figure 1: Solid nodes record completed engineering controls. Dashed nodes remain unresolved. Source: runtime.dot; records: attempts/qemu-v3, admission-v1 and framing-v1.

What the failure controls show

Original full catalog: prime 101 is accepted. N−1/N+1 disabled and only the discriminant catalog deliberately emptied: prime 101 is labeled composite, after two exhaustion messages and a retry. Missing catalog: child exit 139. These controls repeat the prior Docker observations in an independent runtime. They do not show failure on any omitted curve order.

The legacy checker exits zero on all nine fixtures, including rejected and malformed inputs. Strict admission therefore binds root, complete grammar, sentinel, chain order/divisibility, theorem route and checker step count, while guest framing checks completion, ABI and watchdog/process status.

The binding reader is not an independent general mathematical verifier. Earlier order/primality certificates and the affine CM-only 101 check remain separate evidence. The original source-certif2 is an expected rejection; its separately preserved current-reader variant is an expected acceptance.

Workload and runtime

QEMU 11.0.2 on macOS arm64; pinned Alpine 3.20.10 Linux x86 kernel. Fresh one-CPU VM; 256 MiB RAM; no host disks or network. Root-owned read-only inputs; legacy child uid/gid 65534. Child address-space limit 128 MiB, CPU/guest alarm 30 seconds, host watchdog 90 seconds. Actual peak memory and guest CPU consumption were not measured. These limits are not consumption measurements.

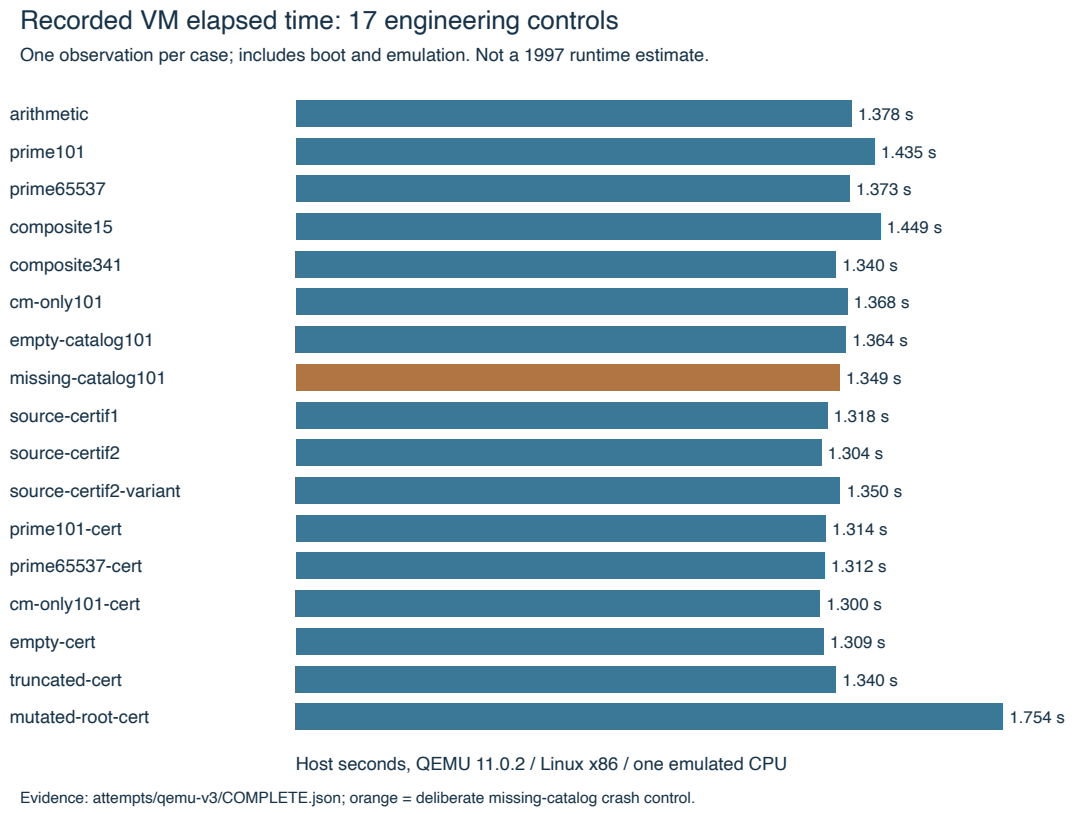


Figure 2: 17 VM host elapsed measurements, one per case. Includes emulation and boot; no historical runtime or speed advantage is inferred. Data: attempts/qemu-v3/COMPLETE.json.

Reproducibility and source identity

Exact commands, input hashes, raw serial, stderr, statuses and host elapsed time are retained in each attempt. Controls ran in the private artifact directory during a Conductor outage; completed records were copied unchanged after scope acquisition recovered. RUNTIME_LOCATION.json binds that location. The earlier qemu-v1 handle disappeared before any recorded compiler/guest result; its observed empty output is retained without inventing an exit reason.

Commands: qemu_controls.py –attempt qemu-v2 –arithmetic-only; qemu_controls.py –attempt qemu-v3; admission_controls.py; framing_controls.py.

runtime-bindings.json inventories original data and executable content hashes. runtime-payloads retains four exact executables in canonical gzip form; all four content digests round-trip. The kernel official bundle checksum matched.

Kernel SHA-256: `ce1f6dad86646efaa980fed238c7ed7e4189fd90c30f99a3788df9c9429cc5c9`

ECPP package SHA-256: `0387e7d180e570d0cf1ff957ed281a46711fb20d50ae0b0669a01a3464d9160f`

The earlier source report documents a 1996 thesis citation to this package filename. It does not authenticate the mirrored bytes as a 1996 capture or establish use by the original generator.

Remaining work and interpretation boundary

The full 21-order cohort is pinned. Its substantive replay still needs the complete frozen execution contract, trial plan, committed Coordinator approval and handoff. The configured Coordinator backend probe reports its API key unset and the backend unusable. No model substitution or approval is asserted. COHORT_PROTOCOL_DRAFT.md records inputs, fixed configuration, censoring and remaining execution work. It is not an approved protocol.

If a later run separates the eleven omissions, it first establishes behavior of this artifact and configuration. Independent replay, wider prediction and original-caller provenance are separate requirements. ECPP uses auxiliary curves over the integer n being proven prime, distinct from the cryptographic curve over F_p. Historical use and any ECDLP advantage remain unestablished.

Evidence: attempts/qemu-v3/COMPLETE.json; attempts/admission-v1/COMPLETE.json; attempts/framing-v1/COMPLETE.json; runtime-summary.json; cohort-inputs.json; runtime-bindings.json; prior nist_seed_ecpp_source_20261024/report.md. The canonical admission diagram was refreshed. Prior report/PDF remain the historical partial snapshot. No mathematical frontier graph changed.